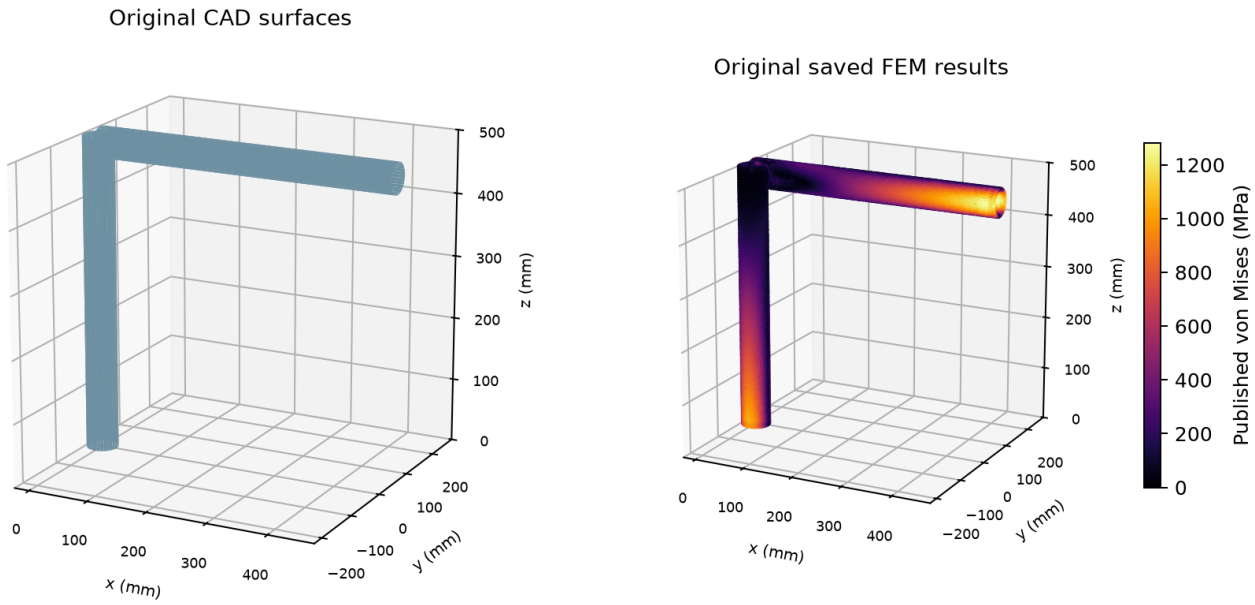
Original contact CAD and FEM results

Status: held, not scored. Original FreeCAD shell-to-shell contact data are preserved. No new benchmark target or engineering problem is generated. The case uses two cylindrical shell surfaces, radius 25 mm and length 500 mm; shell thickness is 0.5 mm.



Published source extrema: maximum displacement 69.691102 mm; maximum von Mises stress 1282.094663 MPa. These values already occur in the native saved Stats array. The figure displays the original saved stress field at 15,404 thickness-expanded shell nodes.

Source setup: E=72,000 MPa, Poisson ratio 0.3, 5,000 N downward point load, frictionless surface-to-surface penalty contact (1,000,000), all translations fixed on the lower z=0 and upper x=475 end edges. The deck uses small-displacement elastic shells; no NLGEOM flag is set.

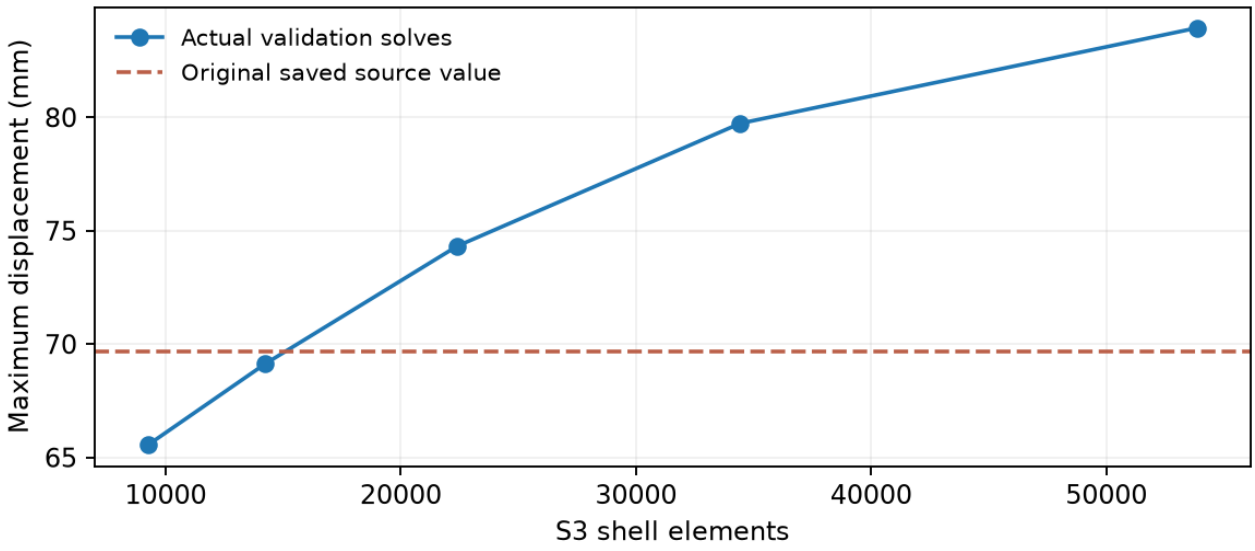
Resolved correspondence: 7,702 input mesh nodes match the native document to 7.04e-11 mm. FreeCAD defines 1 N as 1,000 internal units, so its saved force of 5,000,000 equals the deck load of 5,000 N. Four pinned writer files were verified against their Git blob hashes.

Actual original-deck replay: CalculiX 2.23 gives 69.773736 mm (0.119% above the saved value). Full displacement-field L2 difference is 0.182%; stress-field L2 difference is 2.666%. This is a replay in the same solver family, not independent-formulation validation.

Source: github.com/FreeCAD/FreeCAD, tree 406063b1116a6cb872bbd92ec27c6f425a8d1f98. Original native CAD, deck, scripts and source notices are bundled for the evaluator. Native FCStd contains answers and must not be exposed to candidates.

Mesh sensitivity prevents admission

Five actual remeshed solves retain the source CAD surfaces and setup. The upper cylinder is partitioned in topology so Gmsh can mesh the load vertex; its surface area is asserted unchanged. These new solutions are validation evidence only.



h (mm)	Shell elements	Max u (mm)	Max stress (MPa)
6.4	9236	65.541154	1275.420
5.12	14224	69.123307	1288.850
4.096	22402	74.323073	1626.581
3.2768	34432	79.733876	2214.690
2.6214	53882	83.948853	2533.757

Hold decision: the finest displacement is 20.46% above the saved reference. The last mesh change is 5.02%, and the previous change 6.79%. The existing admission rule requires the last change below 1%, previous below 2%, and a justified reference band within 10%. Peak stress also rises substantially. No acceptance band is widened to admit the case.

Open-shell geometry gate: analytic cylinder axes/radii and Boolean surface coverage replace a solid-volume check. The unchanged original CAD export passes; a 3% scaled test-only copy fails. This standalone validator does not approve normals, setup, mesh, fields or convergence.

Outcome: force units, mesh/result correspondence and shell-geometry checking are resolved. The source remains evaluator-only and unscored because mesh convergence and physical validity are not established. Existing benchmark releases are unchanged. No frontier-model or human expert run is claimed.